Section: **3.8 – COMPOSITE ASSEMBLIES**

Chapter: **3810**

Title: **FLAT STIFFENED COMPOSITE PANELS**

Revision: **0.2**

Date: **10/01/2023**

1 INTRODUCTION

1.1 OBJECT

This document proposes a stress sizing method for the composite skin and stringers of a typical stiffened skin in a lift box, by means of typical approximation of obtaining the internal loads in the structure by either classical stress theories of global Finite Element Models (GFEM) and then calculating the stresses/strains and static margins with analytical (non-FEM) methodologies.

Alternative strategies using detailed Finite Element Models (DFEM) for the structure are not discussed in this document.

Failure modes checked:

- ✓ Skin and stringer strength, considering post-buckling corrections for longitudinal compression and diagonal tension in shear
- ✓ Local buckling of skin, however…
- ✓ Buckling below ultimate loads is permitted consistently with the post-buckling cut-off policy (PBCO, ratio between the ultimate allowable load to the local buckling onset)
- ✓ General/global buckling calculated...
    - as column of the super-stringer (stringer plus effective skin),
    - as panel supported at ribs stations
- ✓ Joint between stringer and skin (only in case the attachment is done by means of fasteners, cocured/bonded joints not covered by this document at the release date of current revision)

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1.2 SCOPE AND LIMITATIONS

Scope and basic limitations:

- ✓ Panels made of fibrous composite materials, flat or with non-significant curvature, stiffened with longitudinal stringers in axial direction

Figure 1. Composite stiffened lift box panel (picture captured from website of Sandvik Coromant)

- ✓ Classical approach of obtaining internal loads in structure by classical stress methods or GFEM's, then check the static margins with closed form and numerical non-FEM methods with the minimum necessary simplification assumptions
- ✓ Laminates according to classical good design rules
- ✓ Each skin subpanel is modelled as a flat shell, with uniform thickness (variable if bonded/cocured stringers) and perimeter supported in stringers/spars
- ✓ The structural element considered in this analysis is the super-stringer, composed by the stringer and adjacent subpanels (halves), effectively supported at ribs stations

See details in following pages…

2 STRUCTURAL DESCRIPTION

The composite stiffened skin, for the purpose of performing its stress and strength analysis, is divided in elementary units. The structural element analysed in this method consists in a piece of the structure contained between two consecutive ribs, and including one stringer and (half) its adjacent skin bays. That is the “super-stringer” element, which is graphically described below.

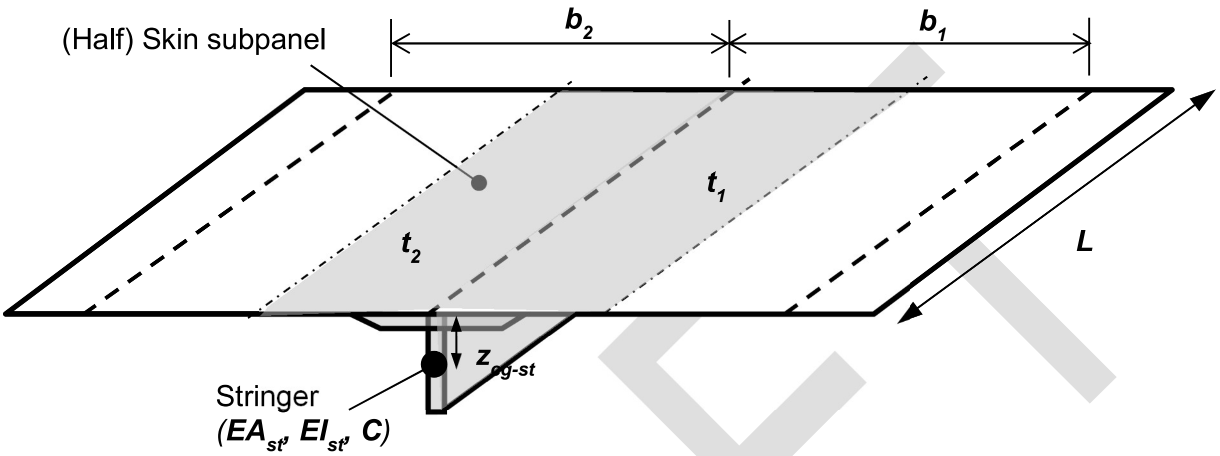

Figure 2. Super-stringer configuration, basic structural element analysed in this method

Some remarks and assumptions in this structural model and method (see details in following pages):

- Axial loads are condensed in the super-stringer cross-section forcing compatibility of longitudinal strain across the super-stringer section. On the other hand, shear and transversal loads are taken specifically for each of the two skin adjacent sub-panels.

- When analysing a complete wing/tail-plane cover, each skin bay is typically checked twice (with slightly different results), as part of two different super-stringers

- For local buckling analysis, the skin subpanels enclosed among ribs and stringers (*L* x *b*) are assumed simply supported at all its edges (*). The over-thickness under cobonded/cocured stiffeners is taken into account in the analysis. Additionally, the torsion stiffness of the stringer can be used as a rotational constraint at lateral edges.

  (*) For recent more “aggressive” designs of panel to ribs attachments (few rivets btw skin and rivets), if the aspect ratio of the skin bays is low, this assumption shall be revised (increasing artificially the length of the skin bay in the buckling calculation)

- General/global buckling is checked by means of two methods:
  - As simply supported stiffened panel composed of (minimum) 6 skin bays between stringers and the 5 stringers contained within that width (“cloning” the configuration of the super-stringer object of analysis)
  - Additionally, the buckling load of the super-stringer between two ribs stations in compression is calculated as column, considering the effective skin width and the stringer crippling cut-off. Classical interaction law can be used with the failure indices found separately (for compression and shear) in order to obtain the static margin.

3 GENERAL STRESS & STRENGTH ANALYSIS PROCESS

The flow diagram below describes roughly the strength analysis process of a super-stringer location. It starts by calculating the pure-allowables, i.e. the allowable loads and strengths corresponding to the possible failure modes in uni-axial loads (laminates strengths, local buckling and crippling loads, column buckling…). Later on, for each load case, the process continues by calculating the stress/strain distribution within the different details (taking into account non-linear post-buckling effects), the corresponding failure indices and finally obtaining the minimum reserve factor (which may require additional iterations for responses that are non-linear with applied loads).

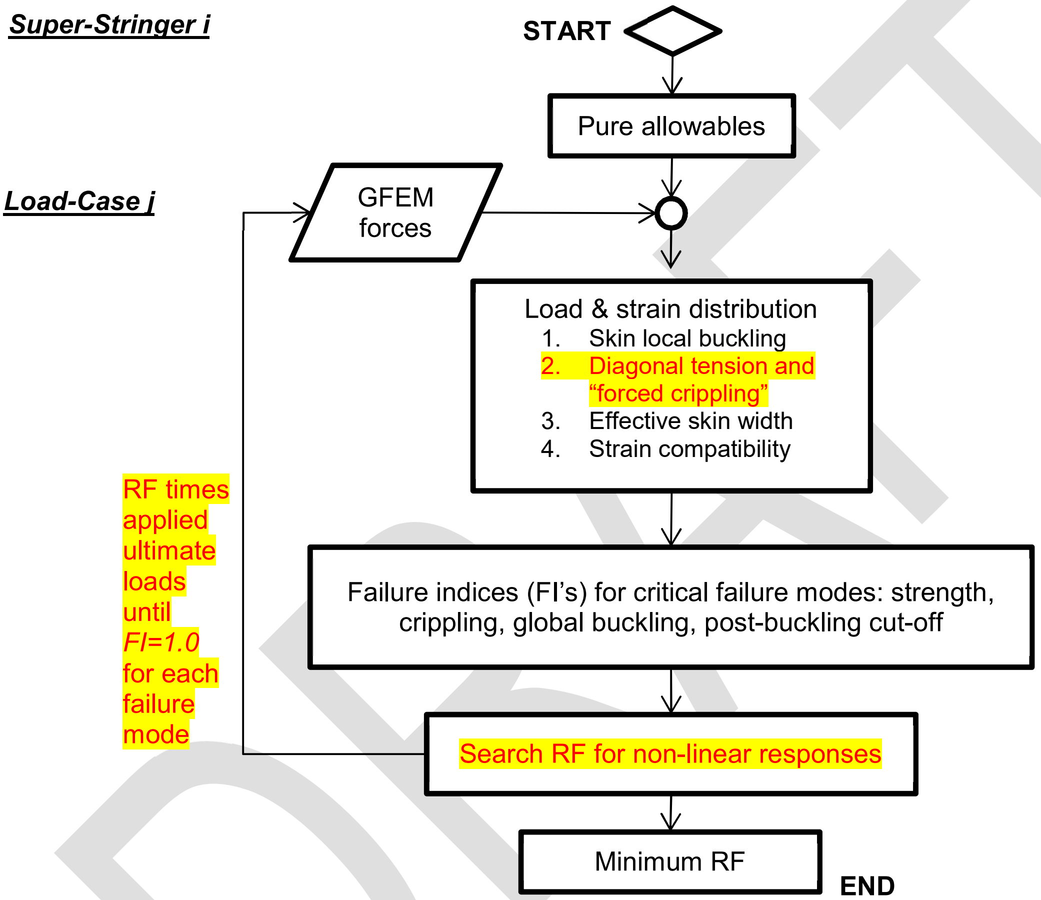

Figure 3. Flow Diagram of the standard strength check process of a super-stringer detail in a composite stiffened panel of an airframe fuselage

4 LOADS, STRESS & STRAIN

4.1 LOADS EXTRACTION FROM GFEM

Typical modelling of the fuselage skin and stringers, in a GFEM, is as follows.

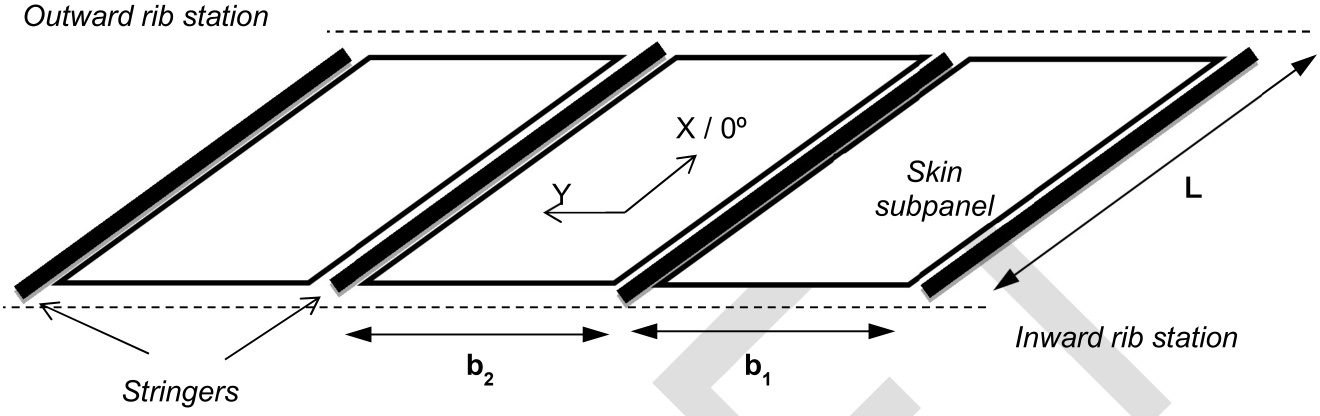

Figure 4. Stress model in GFEM

Notes:

- ✓ There may be more than one FEM element along the distance between support ribs ("L" in figure above). Usually there will be only one element between stringers. Normalization approaches of internal loads for later stress and buckling stability analyses shall typically include averaging and/or taking the elements with maximum and minimum forces.
- ✓ X axis in FEM elements local reference assumed parallel to material stacking 0º direction (otherwise the stiffness properties of the laminates shall be expressed in the analysis reference system of this problem, with X aligned with stringers)

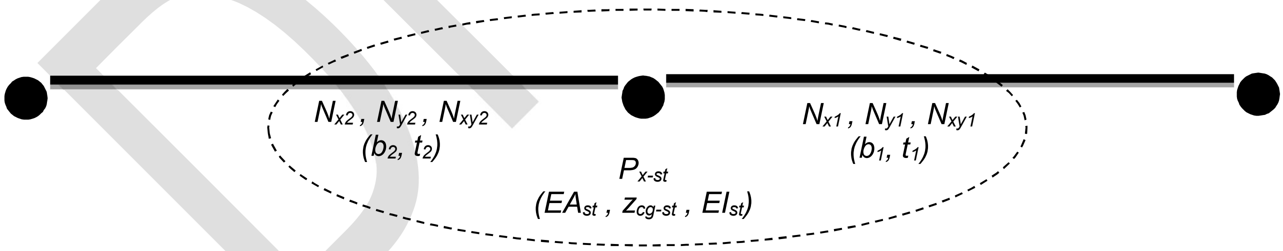

Figure 5. GFEM loads and areas of the super-stringer cross section at a certain location

Internal loads for the structural model, the super-stringer (stringer and semi-skin-subpanels):

- Axial/longitudinal force: $P_x = P_{x\text{-}st} + (b_1 \cdot N_{x1} + b_2 \cdot N_{x2}) / 2$
- Transversal flows: N_{y1} , N_{y2}
- In-plane shear flows: N_{xy1} , N_{xy2}

4.2 STIFFNESS OF ANALYSIS LOCATION

4.2.1 Super-stringer cross-section stiffness

Considering discussion about the effective width of skin, the centre of gravity and inertia properties of the super-stringer cross section are calculated as follows:

$w_1 = b_1 / 2$ $\qquad w_2 = b_2 / 2$

$t_{pad} = max(t_1, t_2)$ $\qquad E_{pad} = E_{x1}$ if $t_1 > t_2$,

$E_{pad} = E_{x2}$ otherwise (*)

$$EA = EA_{st} + E_{x1}\cdot t_1\cdot(w_1 - b_{pad}/2) + E_{x2}\cdot t_2\cdot(w_2 - b_{pad}/2) + E_{pad}\cdot t_{pad}\cdot b_{pad}$$

$$z_{cg} = [\ EA_{st}\cdot(t_{pad} + z_{cg\text{-}st}) + 0.5\cdot E_{x1}\cdot t_1^2\cdot(w_1 - b_{pad}/2) + 0.5\cdot E_{x2}\cdot t_2^2\cdot(w_2 - b_{pad}/2) + 0.5\cdot E_{pad}\cdot t_{pad}^2\cdot b_{pad}\ ] / EA$$

$$EI = EI_{st} + EA_{st}\cdot(t_{pad} + z_{cg\text{-}st} - z_{cg})^2 + E_{x1}\cdot t_1\cdot(w_1 - b_{pad}/2)\cdot(t_1 - z_{cg})^2 + E_{x2}\cdot t_2\cdot(w_2 - b_{pad}/2)\cdot(t_2 - z_{cg})^2 +$$
$$+ E_{pad}\cdot t_{pad}\cdot b_{pad}\cdot(t_{pad} - z_{cg})^2$$

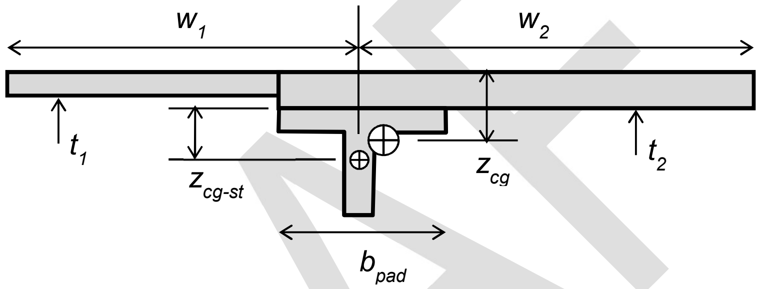

Figure 6. Super-stringer cross section

(*) Note the "pad" laminate is assumed that of the skin laminates with greater thickness. A dedicated, thicker, pad laminate could be considered with the same formulation

4.2.2 Super-stringer simplified idealization as a laminate

Idealizing the super-stringer assembly, i.e. the sum of the stringer plus the skin, as a fictitious laminate with the stiffness equivalent to the real assembly, is an approach of utility for global buckling calculations, proposed historically by several authors. This topic is discussed later in this chapter and in chapter 3453 "GLOBAL BUCKLING OF FLAT COMPOSITE STIFFENED PANELS".

4.3 STRESS & STRAIN DISTRIBUTION

Strain compatibility and loads equilibrium require that:

- ✓ The axial strain ε_x shall be uniform across the section
- ✓ The membrane load flows N_y and N_{xy} shall be uniform across the strips of each subpanel (raw skin, skin pad, …)
- ✓ Contributions of the different sub-elements of the super-stringer sum the total axial loads…:
  - total axial force $\qquad P_x = \Sigma(N_{x\text{-}i}\cdot b_i) + P_{strg}$

- Hook's law in stringer $P_{strg} = (EA)_{strg} \cdot \varepsilon_x$
- …in skin&pad laminates $[A]_i \cdot \{\varepsilon\}_i = \{N\}_i$

…with $\{\varepsilon\}_i = (\varepsilon_x, \varepsilon_{y-i}, \gamma_{xy-i})$, $\{N\}_i = (N_{x-i}, N_y, N_{xy})$

These conditions shall permit to get the stresses/strains at any point and detail of the section (note that, in the GFEM, typically details as flanges and pads will be condensed into a 1D element without the required sensitivity for certain analyses).

5 FAILURE MODES

5.1 SKIN FAILURE

5.1.1 Local buckling

Forces in skin elements shall be normalized into uniform in-plane/membrane loads for each subpanel (skin bay) delimited by 2 adjacent stringers and 2 consecutive ribs, which will be considered effective supports in this analysis. For typical subpanels of relatively high aspect ratios (ribs spacing more than twice the stringer pitch) the recommended approach shall be averaging in width (chord direction) and taking the elements forces values of greatest compression and shear.

Once internal loads at each subpanel have been normalized, buckling load levels shall be calculated as per chapter §3450 "PLATE/SHELL BUCKLING".

5.1.2 Strength

Strain distribution in the super-stringer assembly is initially available per calculation method described in previous section.

Post-buckling in compression, in shear and in combined loads

For cases with axial compression, the strain distribution may need to be recalculated if local buckling of the skin is permitted below ultimate loads. Revised strain distribution shall be obtained per method in chapter §3485 "EFFECTIVE SKIN WIDTH"

For cases with skin in shear and local buckling below ultimate loads, strain distribution in skin may need to be recalculated using diagonal tension theory as per defined in chapter §3460 "SHEAR WEBS". This approach shall force additional compression forces in stringer which shall have to be added to the internal loads calculation loop.

Skin Failure

Provided the strain state of the skin is known, taking into account post-buckling effects if any, the static margin for skin strength shall be calculated according to failure criteria set by chapter §3403 "PLAIN STRENGTH OF COMPOSITE LAMINATES" or, more typically if the part has damage tolerant requirements, those by chapter §3405 "RESIDUAL STRENGTH AFTER BVID IMPACT".

As the response of the stress/strain with load is non-linear if local buckling is permitted (see discussion about effective skin width and diagonal tension), the reserve factor shall be preferably calculated as times the applied loads causing the failure index at skin equal to 1.0.

5.1.3 Post-buckling policy

The capacity to exceed the loads at which skin onset occurs shall be limited as a design policy, then mitigating the risks of undesired failure derived by the complexity of the post-buckling behaviour (even previously defined methodology accounted post-buckling effect up to some extent). This post-buckling policy is established by means of a post-buckling-cut-off factor *PBCO* , which is defined as the ratio between allowable loads and the local buckling onset of the skin

$$PBCO = allowable\_load / buckling\_load$$

Guidance for selecting design values for this parameter is given in chapter §3468 “POST-BUCKLING OF COMPOSITE PANELS”, Each project may define alternative post-buckling policies, however, provided they are adequately substantiated (typ. by dedicated testing).

5.2 STRINGER FAILURE

5.2.1 Strength

Plain strength or residual strength after impact shall be checked vs the axial strain in the stringer.

In most cases, the general approximation model for residual strength after transverse impact in “big” panels shall not be applicable and special design values shall be needed (eg. compression allowable strains after top/edge impact in tee or L shaped stringers). See chapter §1422 “RESIDUAL STRENGTH AFTER EDGE IMPACT IN COMPOSITES” for further guidance.

5.2.2 Local buckling and crippling

The crippling strength of the stringer shall be calculated according to method described in chapter §3430 “LOCAL BUCKLING AND CRIPPLING OF BEAMS/STRINGERS”.

Beyond its use in the strength check of the stringer in compression, the crippling strength of the stringer shall serve also as cut-off value for the column buckling load used in global stability analyses (Johnson-Euler curve).

As discussed for the skin, static margins for both failure modes shall be calculated preferably as times the load cause failure index equal to 1.0, taking into account non-linearities (effective skin width function of the stress).

5.3 STRENGTH OF SKIN-STRINGER JOINT

The load transfer through joint between the stringer and the skin shall normally not be a critical if standard design rules were followed, with some remarkable exceptions as stringer run-out details (out of the scope of this chapter). By-pass axial load can be critical however if the joint is riveted.

For typical super-stringer location, loads transferred through the skin-stringer joint can be extracted as explained in chapter §3215 “FORCES EXTRACTION FOR ANALYSIS OF BOLTED JOINTS”. The shear load sustained by this joint shall be equal to the change in stringer axial load divided by the increment in length. With typical stringer idealizations using 1D elements:

$$N_s = \Delta P_x / \Delta L$$

5.3.1 Fastened Joint

Only in case the stringer is joined to the skin by means of rivets. The shear load at each rivet if attachment is done with *n* rows of rivets at pitch *p* will be:

$$Q = N_s \cdot p / n$$

Fastened joint failure

The method to calculate the static margin for the fastener joint is presented in chapter §3620 “STRENGTH OF FASTENED JOINTS IN COMPOSITES”. By-pass loads in skin and stringer foot are also required as input in this analysis, in addition to the rivet shear loads.

Inter-rivet buckling

Inter-rivet buckling stress is used as cut-off value, in addition to the crippling load of the stringer, in effective skin calculation. The inter-rivet buckling stress has to be obtained per method in chapter §3480 “INTER-RIVET BUCKLING”. Notice inter-rivet buckling is not considered failure and has not a reserve factor associated.

5.3.2 Bonded (or cocured) Joint

The shear load N_s to be sustained by this joint shall be compared with its allowable, which has to be calculated for the particular materials, laminates and geometry (foot width).

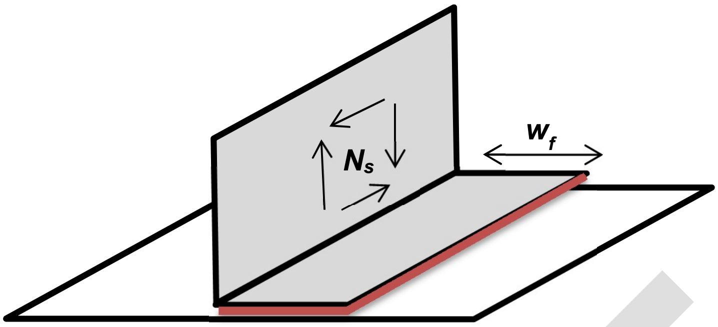

Figure 7. Sketch of a stringer foot bonding to skin

See chapters §3710 "STRESS ANALYSIS OF A BONDED JOINT" and §3730 "FAILURE CRITERION" for details about the calculation of the strength of this adhesive/cohesive attachment as in-plane-shear lap joint.

5.4 GLOBAL BUCKLING OF THE SUPER-STRINGER

The configuration of the whole super-stringer element is needed for computing the static margin vs general/global stability, at the analysis locations. Two complementary methods are used for purpose, discussed in more detail in chapter §3453 "GLOBAL BUCKLING OF FLAT COMPOSITE STIFFENED PANELS"…

5.4.1 Global Buckling as column-panel

Column Buckling

The column buckling stability of the beam composed by the stringer plus the corresponding effective skin, simply supported at frames stations, under compression, shall be checked as per method defined in chapter §3440 "COLUMN BUCKLING OF COMPOSITE BEAMS". The stringer crippling stress (and the inter-rivet buckling if applies) shall be used as cut-off value for the Johnson-Euler buckling calculation.

In the column buckling calculation, the rotational restriction at ribs supports is controlled by means of the fixity coefficient *C*. For conventional designs, with ribs including mouse-holes at stringers crossings, and not attached directly to stringers, *C* shall be made equal to 1.0 by default (thus, simply supported condition, see discussion in Ref. 4). The use of values higher than 1.0 should be properly justified (typ. by sufficiently representative tests or DFEM).

Shear buckling

The allowable load of the super-stringer element in in-plane shear can be computed from any of the methods available high-fidelity (Rayleigh-Ritz or FEM) or approximated, as the "equivalent plate method" suggested in §3453.

Buckling in combined loads

With the failure indices in compression (vs column buckling) and in shear, the static margin will be calculated with the interaction law proposed in chapter §3450 "BUCKLING OF COMPOSITE PLATES"…

$$RF = 2 / [\, R_c + (R_c^2 + 4 \cdot R_s^2\,)^{0.5}\,]$$

5.4.2 Global Buckling as panel with stable cross-section

By the most adequate method of those discussed in §3453 , a fictitious stiffened panel is analysed, this panel consisting in the super-stringer element of interest repeated in the direction of the chord a sufficient number of times to get a representative configuration for a global stability analysis. The limits of the panel in direction of the span // X, are the ribs stations.

The configuration of this panel across the width to be set by these conditions:

- ✓ The minimum number of skin bays considered is 6
- ✓ The total width of the panel shall be equal or greater than its length (*L*), i.e. the ribs distance

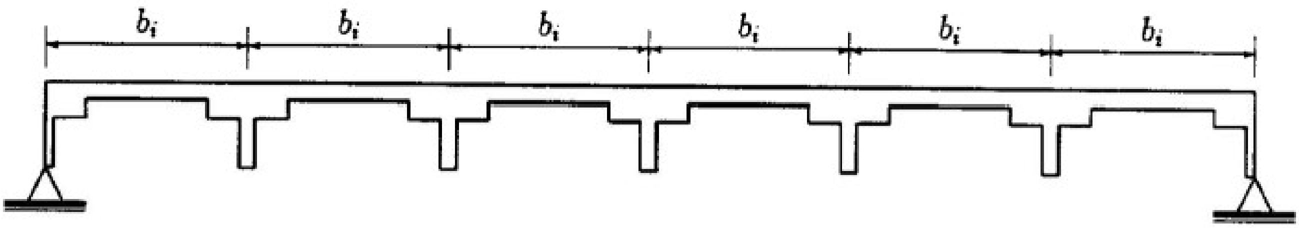

Figure 8. Scheme of the stiffened panel considered in global buckling analysis

The perimeter of this stiffened panel shall be considered simply supported for the buckling analysis. Anyway, when the method used is suitable, the torsional stiffness of the stringer shall be used as rotational constraint at the lateral sides of the panel.

The reserve factors for global buckling failure obtained by these two methods (5.4.1 and 5.4.2) will not be equivalent. The minimum of the two shall be considered the critical (in absence of a more accurate prediction from a method of higher fidelity, e.g. detailed FEM simulation of the whole lift box).

6 APPENDICES

ABBREVIATIONS AND SYMBOLS

Symbol	*Units*	*Definition*
[A]	[N/mm]	In-plane stiffness matrix of the skin laminate
[B]	[N]	In-plane / bending coupling matrix of the skin laminate
b	mm	Width of the skin (distance between stringers)
b_b	mm	Buckling width of the skin, distance between nearest walls of stringers in case they are omega shaped
BL	-	Buckling Level (Buckling_Load / Applied_Load)
BVID	-	Barely Visual Impact Damage. Impact damage in the threshold of detectability.
C	-	End fixity coefficient for column buckling analysis
C_c , C_s	-	Coefficients for calculation of post-buckling margin policy
CAI	-	Compression residual strength of laminate after BVID impact, as allowable strain
[D]	[Nmm]	Bending stiffness matrix of the skin laminate
EA	N	Axial inertia of the super-stringer (stringer plus effective skin)
EA_{st}	N	Axial inertia of the stringer (without skin)
EI	Nmm^2	Bending moment of inertia of the super-stringer
EI_{st}	Nmm^2	Bending moment of inertia of the stringer (without skin)
ε	-	Strain in a composite laminate, at certain direction (typ. 0º / X , 90º / Y or ±45º)
FI	-	Failure Index, defined at detail/failure-mode level as the applied stress/strain/load divided by the allowable stress/strain/load
FRP	-	Fiber Reinforced Polymer
GFEM	-	Global Finite Element Model
L , a	mm	Length of the stringer, and of the adjacent skin subpanels
LL	-	Limit Load
m	-	Number of semi-waves in longitudinal X- direction of the buckling shape
n	-	Number of semi-waves in transversal Y-direction of the buckling shape, or number of rivets rows, per scope
Ns	N/mm	Shear load transferred by a longitudinal joint
N_y	N/mm	Axial transversal flow
N_{xy}	N/mm	Shear flow in skin
N_{xb}	N/mm	Buckling load of the skin subpanel in longitudinal compression
N_{yb}	N/mm	Buckling load of the skin subpanel in transversal
N_{xyb}	N/mm	Buckling load of the skin subpanel in shear
p	mm	Rivets pitch
P	N	Load force

PBCO	-	Ratio specifying the ultimate allowable load exceeding the buckling onset: PBCO = [Allowable_Load] / Buckling_Load
P_{col}	N	Column buckling load of the super-stringer in longitudinal compression
P_x	N	Axial longitudinal force in the whole section of the super-stringer
Q	N	Rivet shear load
RF	-	Reserve Factor (Allowable_Load / Applied_Load)
R_x	-	Failure index in X-load (X_Applied_Load / X_Allowable_Load)
R_{xy} , R_s	-	Failure index in shear XY load (XY_Applied_Load / XY_Allowable_Load)
R_y	-	Failure index in Y-load (Y_Applied_Load / Y_Allowable_Load)
R_c	-	Failure index in bi-axial loads, shear excluded (Rx+Ry if linear interaction)
strg	-	As subscript, means applicable to the stringer
t	mm	Thickness of the skin
TAI	-	Tension residual strength of laminate after BVID impact, as allowable strain
UL	-	Ultimate Load
w	mm	Effective semi-width, i.e. skin width adjacent (at one side) to the stringer support that is still sustaining compression after buckling
z_{cg}	mm	Distance from outer skin face to the super-stringer center of gravity
$z_{cg\text{-}st}$	mm	Distance from the inner skin face to the stringer center of gravity (without skin)

REFERENCES

	Document Reference	*Issue*	*Title*
Ref. 1	TEA-T-NT-210048	Ed. B	Manual Teórico del Programa ARPA
Ref. 2	NT-00-ECE-044	Ed. A	Procedimiento de análisis y diseño de revestimientos de superficies sustentadoras”
Ref. 3	[Christos Kassapoglou]	2nd Ed.	Design and Analysis of Composite Structures
Ref. 4	TE-F9-MM-220035	Is. A	BUCKLING ANALYSIS METHODOLOGIES FOR THE STIFFENED PANELS OF THE F10X TAIL PLANES

SOFTWARE

Program	***Version***	***Description***
ARPA (Airbus DS)	5.2	Strength analysis of flat composite stiffened panels
STRENGTH2000 (Airbus DS)	X.X	Detailed sizing of metallic and composite aircraft structures

Table 2. *Software programs referenced in this chapter*

Remark: The methodology described in this chapter is generally aligned with the analysis process performed by ARPA software. The software STRENGTH2000 (S2K) includes already some of the elementary methods discussed along the chapter, and is to be completed with the remaining and the whole calculation process of the stiffened panel, at its time.

RECORD OF REVISIONS

Revision Date	***Authors***	***Change Reason***	***Pages/Sub-chapters affected***
0.1 30/09/2022	G. Baños	First issue	All pages

Table 3. *Record of Revisions: authors, change reasons and sections/pages affected*

APPROVAL SIGNATURES TABLE

Dpt. \ Site	***Getafe***	***Manching***
Stress Methods (TEPAP-TL4)	G. Baños 30/09/2022	F. Glock DD/12/2022

Table 4. *Approval signatures table for last revision of this chapter*